INCH-POUND

MIL-C-85049/2A
30 November 1990
SUPERSEDING
MIL-C-85049/2
16 January 1980
MS3057E
10 May 1982

MILITARY SPECIFICATION SHEET

CONNECTOR ACCESSORIES, ELECTRICAL, BACKSHELL, ENVIRONMENTAL, CABLE SEALING, STRAIGHT, CATEGORY 1C (FOR MIL-C-5015 SOLDER TYPE, V THREAD OF MS310X CLASSES A, B, C OR K CONNECTORS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-85049.

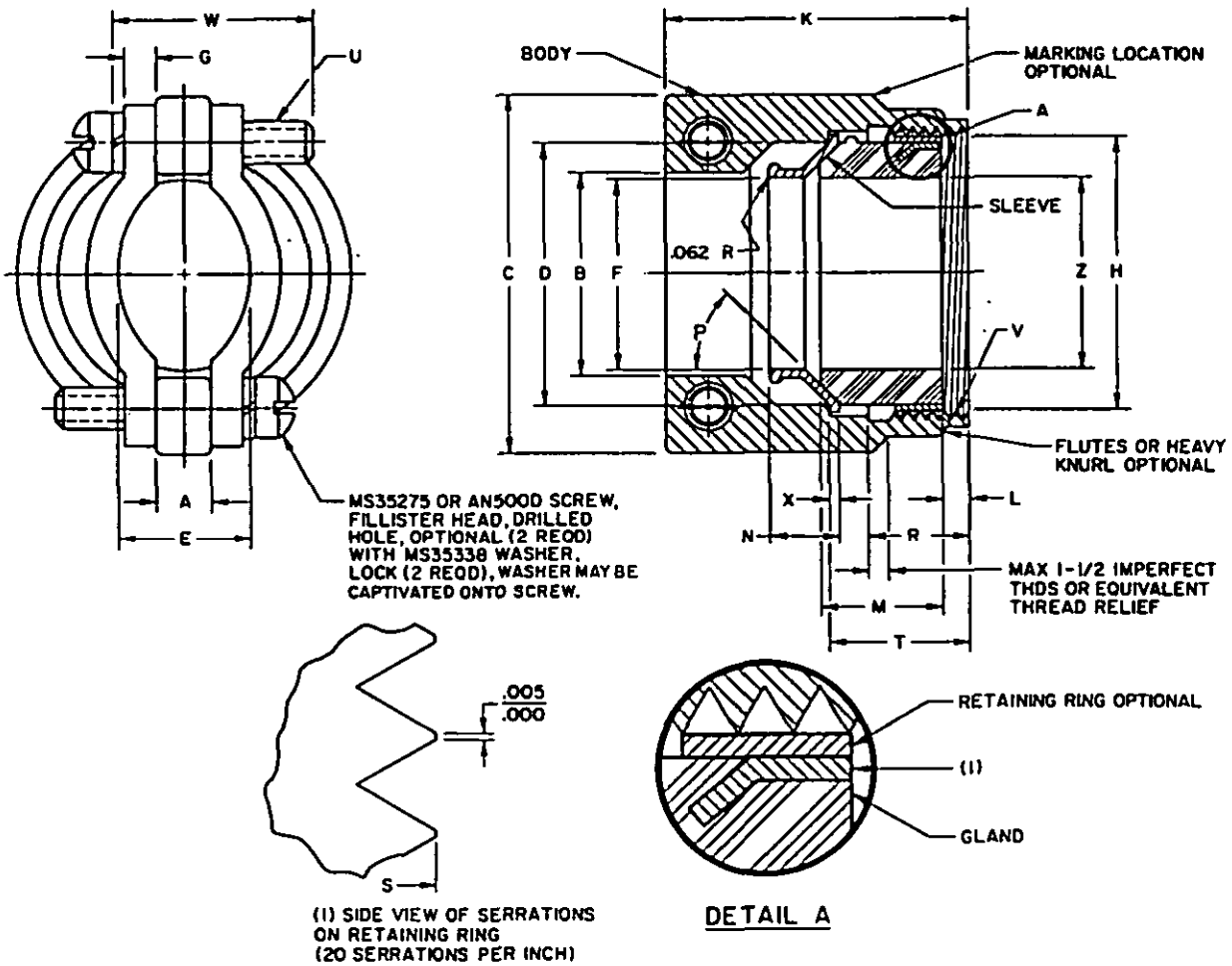


FIGURE 1. Configuration and dimensions.

AMSC N/A FSC 5935

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TABLE I. Dimensions.

Dash No.	Connector Shell Size (Ref) 1/	A +.031 -.016	B Min Cable Clearance	C Max	D +.000 -.020 Dia Gland	E Max Closed	F +.020 -.000 Dia	G ±.016	H +.000 -.020 Dia	K Max	L Min	M ±.031	N	P Min	R Min
-3C	8S and 10S	.188	.239	.812	.389	.217	.231	.125	.441	1.375	.156	.473	.254	20°	.531
-4C	10SL, 12S and 12		.317	.937	.510	.296	.315		.560					30°	
-6C	14S and 14		.443	1.062	.627	.390	.440		.676						
-8C	16S and 16		.551	1.188	.744	.434	.515		.800						
-10C	18		.630	1.312	.870		.614		.926	1.437		.537	.281	30°	
-12C	20 and 22	.250	.771	1.562	1.037	.596	.738	.156	1.107			.625	.320		
-16C	24 and 28		.959	1.750	1.286		.926		1.357	1.562		.662	.344		
-20C	32	.312	1.271	2.250	1.590	.858	1.200		1.670	1.812			.406	35°	.562
-24C	36		1.396	2.375	1.844		1.363	.188	1.920	2.062		.940			.781
-28C	40		1.646	2.625	2.083	1.107	1.611		2.162						
-32C	44	.375	1.896	2.812	2.332	1.421	1.865		2.412	2.188	.188	1.047	.500		

Dash No.	T	U Thread	Fillister Head Screw AN500D Dash No.	Fillister Head Screw MS35275 Dash No.	Screw Size	V Thread	W Min	X	Z Dia Approx (Gland) Free	Z Dia Approx (Gland) Closed	Bushing (Ref)	Max Cable Entry	MS35338 Washer, Lock Dash Number
-3C	.641	.138-32UNC-2A	6-8	230	(6-32)	.500-28 UNEF-2B	.469	.025 +.010 -.003	.219	.031	MS3420-3A	.219	136
-4C			6-10	231		.625-24 UNEF-2B	.594		.312	.094	MS3420-4A	.312	136
-6C			6-10	231		.750-20 UNEF-2B			.438	.250	MS3420-6A	.438	136
-8C			6-12	232		.875-20 UNEF-2B	.719		.531	.312	MS3420-8A	.515	136
-10C	.672		6-12	232		1.000-20 UNEF-2B			.625	.375	MS3420-10A	.614	136
-12C	.719	.164-32UNC-2A	8-14	248	(8-32)	1.188-18 UNEF-2B	.844	.032 +.012 -.003	.750	.500	MS3420-12A	.738	137
-16C			8-18	NA		1.438-18 UNEF-2B	1.062		.938	.594	MS3420-16A	.926	137
-20C	.797	.250-20UNC-2A	416-20	284	(1/4-20)	1.750-18 UNS-2B	1.187	.051±.005	1.250	.938	MS3420-20A	1.200	139
-24C	1.016		416-22	NA		2.000-18 UNS-2B	1.312		1.375	.969	MS3420-24A	1.363	139
-28C			416-22	NA		2.250-16 UN-2B			1.625	1.138	MS3420-28A	1.611	139
-32C	1.172		416-22	NA		2.500-16 UN-2B		.064±.005	1.875	1.375	MS3420-32A	1.865	139

1/ Clamp mates with connector shell size listed in this column.
NA = Not Available

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INCH	mm	INCH	mm	INCH	mm
0.003	0.08	0.510	12.95	1.172	29.77
0.005	0.13	0.515	13.08	1.187	30.15
0.010	0.25	0.531	13.49	1.188	30.18
0.012	0.30	0.537	13.64	1.200	30.48
0.013	0.33	0.551	14.00	1.250	31.75
0.016	0.41	0.560	14.22	1.271	32.28
0.025	0.64	0.562	14.27	1.286	32.66
0.031	0.79	0.594	15.09	1.312	33.32
0.032	0.81	0.596	15.14	1.357	34.47
0.051	1.30	0.614	15.60	1.363	34.62
0.0625	1.59	0.625	15.88	1.375	34.93
0.094	2.39	0.627	15.93	1.396	35.46
0.125	3.18	0.630	16.00	1.421	36.09
0.138	3.51	0.641	16.28	1.437	36.50
0.156	3.96	0.662	16.81	1.438	36.53
0.164	4.17	0.672	17.07	1.562	39.67
0.188	4.78	0.676	17.17	1.590	40.39
0.217	5.51	0.719	18.26	1.611	40.92
0.219	5.56	0.738	18.75	1.625	41.28
0.231	5.87	0.744	18.90	1.646	41.81
0.239	6.07	0.750	19.05	1.670	42.42
0.250	6.35	0.771	19.58	1.750	44.45
0.254	6.45	0.781	19.84	1.812	46.02
0.281	7.14	0.797	20.24	1.844	46.84
0.296	7.52	0.800	20.32	1.865	47.37
0.312	7.92	0.812	20.62	1.875	47.63
0.317	8.05	0.844	21.44	1.896	48.16
0.320	8.13	0.858	21.79	1.920	48.77
0.344	8.74	0.870	22.10	2.000	50.80
0.375	9.53	0.926	23.52	2.062	52.37
0.389	9.88	0.937	23.80	2.083	52.91
0.390	9.91	0.938	23.83	2.162	54.91
0.406	10.31	0.940	23.88	2.188	55.58
0.434	11.02	0.959	24.36	2.250	57.15
0.438	11.13	0.969	24.61	2.332	59.23
0.440	11.18	1.000	25.40	2.375	60.32
0.441	11.20	1.047	26.59	2.412	61.26
0.443	11.25	1.062	26.97	2.500	63.50
0.469	11.91	1.107	28.12	2.625	66.68
0.473	12.01	1.138	28.91	2.812	71.42
0.500	12.70				

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REQUIREMENTS:

1. Design and construction:

   Dimensions and configuration: See figure 1 and table I.

2. Lubricant, VV-P-236, shall be applied to threads and surfaces of sleeve.

3. Optional retaining ring shall be bonded to gland.

4. For sizes -3 through -10, serrations shall be covered with rubber. For sizes larger than -10, serrations may be covered with rubber. Depth of rubber shall not exceed .013.

5. Material and finish: See table II.

6. Qualification: See MIL-C-85049, Category 1C. Information pertaining to qualification of products may be obtained from the Commanding Officer, Naval Avionics Center, Attention: Code B/444, 6000 East 21st Street, Indianapolis, IN 46219-2189.

7. Fluid immersion requirements not applicable.

8. Part number: See example below:

9. Military designator — M85049/
   Specification sheet — 2
   Dash number (see table I) — -3C

8. Supersession data:

MS3057C - * * M85049/2 - *

Dash Number

Finish

Old	New
A	W
C	W
N	W

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TABLE II. Material and finish.

	Material	Finish
Body	Aluminum alloy in accordance with MIL-C-85049.	1/ W
Screws and washers	Corrosion resistant steel in accordance with MIL-C-85049.	Passivate - per QQ-P-35
Sleeve	Aluminum alloy in accordance with MIL-C-85049.	In accordance with MIL-C-5541. The portion of the sleeve in contact with the body shall be coated with polytetrafluoroethylene resin for sizes 3 through 10, optional.
Gland	Rubber AMS 3209, durameter A, 70 ± 5	None
Retaining ring optional	Brass SAE 70, hard temper only	None
Clamping bars	Aluminum in accordance with MIL-C-85049.	1/ W

1/ W finish is not for use in space application.

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are based upon 1.00 inch = 25.4 mm and shall be in accordance with ASTM E380, paragraph 4.5, Method A and ASTM E29, rounding-off method.
3. Unless otherwise specified, tolerances shall be as follows: Decimals ±.016, angles ±2°.
4. Dimensions apply after plating.
5. Dimensions and tolerances in accordance with ANSI Y14.5.
6. Remove all burrs and edges.
7. For additional closedown of the Z dimension, bushings MS3420-3A through MS3420-40A may be used. Diameter Z shall be free of obstructions.
8. Bushings (MS3420) are not supplied with the backshell and must be procured separately.

Revision letters are not used to denote changes due to the extensiveness of the changes.

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Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - AR, MI
Navy - EC
Air Force - 11, 99
DLA - ES

Preparing activity:
Navy - AS
(Project No. 5935-3729-02)

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